

Task Description The system is designed to streamline the seat allocation process in university admissions
Objective To develop a fair and transparent ranking system based on predefined criteria.
Submission The project will be submitted in the form of a detailed report, along with the source code and a presentation.

Introduction

- The system calculates rank for a set of students based on their cutoff and caste allocates seats for them based on their choice filling preferences.

Application: It is used during admission process like university admissions, counselling systems, etc.

It has three subtasks

1. New registration for user – Getting information of a new user like marks, name, etc.
2. Rank Generation – Generates rank for students by sorting of New_Value (based on caste and cutoff)
3. Seat Allocation – Based on available seats and choice filled by candidates.

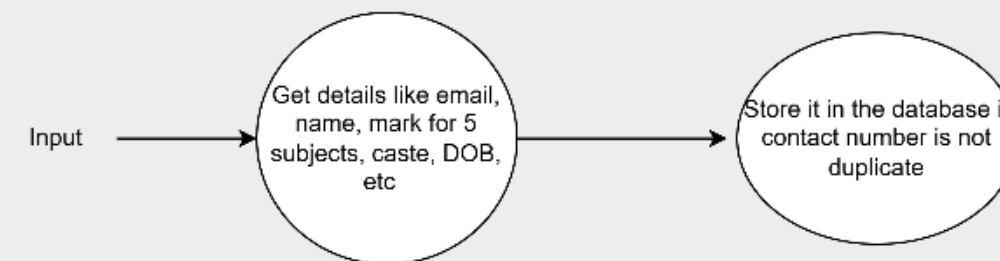
Important URL'S

1. adlogin/ - URL for directing to admin login.
2. adlogin_check/ – URL for verifying the login credentials of the admin when logged in as admin.
3. viewusers/ - URL for displaying all the candidates registered for the allotment.
4. editprofile/ - URL for displaying detailed information of a single candidate.
5. login/ - URL for existing user login.
6. userreg/ - URL for getting details of new user.
7. insertuser/ - URL for new user registration.
8. verify_otp/ - URL for verifying OTP of new user.
9. login_check/ - URL for verifying the login credentilas of the user.

Methodology

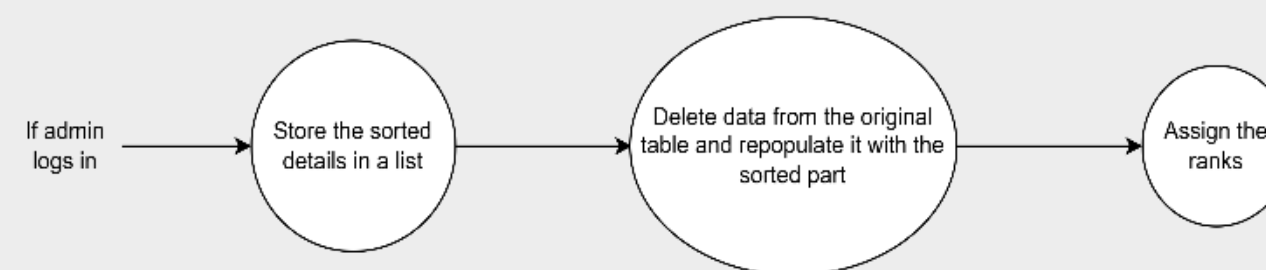
Sub task 1 : New user registration

To store the details of a new student.



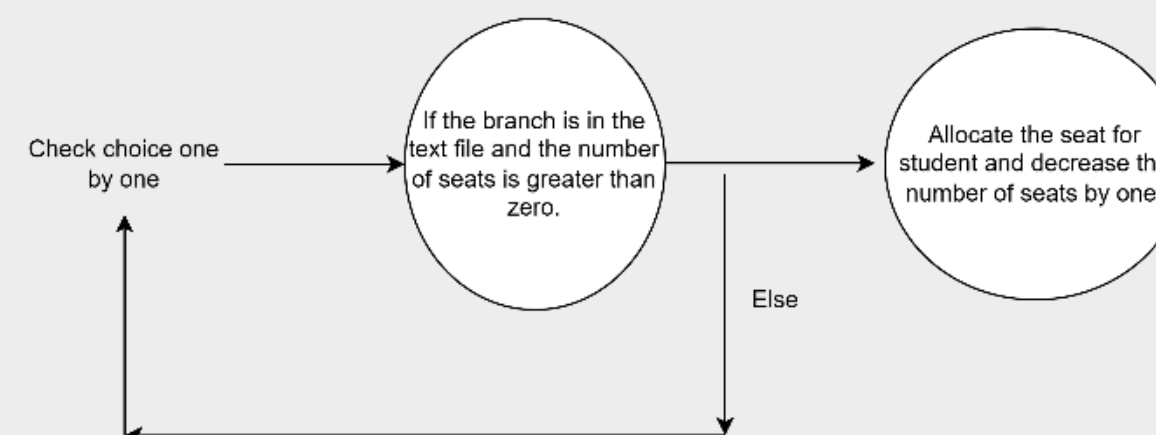
Sub task 2: Rank generation

To generate rank for existing students.



Sub task 3: Seat Allocation

To allocate the seats based on choice



Other functions

1. Login Module - This module shows a user login and registration process.
2. New Registration Module: This module describes a candidate registration process. It collects the candidate's name and details.
3. Cutoff Calculation: This module calculates the cutoff and new value.
4. Rank Generation: This module describes the process of generating a rank list for candidates. Upon admin login, candidate details are stored in a list and sorted by new_value in descending order.
5. Display result: This module outlines the process for printing candidate details.
6. Seat Allocation: This module allocates seats to students based on their choice filled and available seats and their marks.

NOTE:

In case of same New_value for two or more candidates the preference is for maths, physics, chemistry, total, birthyear, name.

Inferences

Suitable for Practical application : If reflects the original seat allocation software as our software also allots seat based on merit.

Enhanced Fairness: By basing rankings on predefined criteria such as cutoff marks and caste, the system ensures a fairer distribution of seats compared to manual